SOFTWARE REQUIREMENTS SPECIFICATION

SRS DOCUMENT

Corvit AI Chatbot

AI-Powered Student Guidance and Information System

Document Version:	1.0
Date:	September 2026
Project Type:	AI-Powered Web Application
Repository:	`corvit-ai-chatbot`

Prepared By

Muhammad Zain Tariq

Technology Stack

HTML • CSS • JavaScript • Tailwind CSS • GPT-OSS-120B • RAG • Online Search

A professional AI-powered platform for Corvit course discovery, student assistance, information retrieval, and career guidance.

Document Information

Project Name	Corvit AI Chatbot
Document	Software Requirements Specification
Version	1.0
Prepared By	Muhammad Zain Tariq
Date	September 2026
Repository	`corvit-ai-chatbot`

Revision History

Version	**Date**	**Description**	**Author**
1.0	Sep. 2026	Initial SRS	Muhammad Zain Tariq

Abstract

The **Corvit AI Chatbot** is an AI-powered web-based student assistance system designed to simplify access to information related to Corvit Systems. Students and prospective learners may need information about courses, certifications, instructors, schedules, facilities, NAVTTC programs, and career opportunities.

Traditional websites mainly provide static information and require users to manually search through different pages. The proposed system introduces an intelligent conversational interface that allows users to ask questions using natural language.

The system combines a professional responsive website with an AI chatbot powered by **GPT-OSS-120B**. A structured Corvit knowledge base and retrieval mechanism are used to provide relevant institution-specific information. Online search may also be used when current external information is required.

The objective is to provide a centralized, interactive, accessible, and intelligent platform for student guidance and institutional information.

Contents

1 Introduction

1.1 Purpose

This Software Requirements Specification defines the requirements of the Corvit AI Chatbot. It describes the system's purpose, scope, functional requirements, non-functional requirements, architecture, user interactions, security requirements, testing requirements, and future enhancements.

The document is intended for project developers, supervisors, testers, stakeholders, and future maintainers.

1.2 Problem Statement

Students and prospective learners frequently need information about training programs, certifications, schedules, instructors, infrastructure, NAVTTC opportunities, and career pathways.

In a conventional website environment, this information may be distributed across different pages or communication channels. Users may therefore spend significant time searching for information or contacting staff for common questions.

The major problems are:

- Difficulty finding relevant information quickly.
- Static websites cannot understand natural language queries.
- Students may not know which course best matches their goals.
- Common questions may require repeated manual assistance.
- Information may exist in multiple sources.
- Users lack a centralized conversational guidance system.

1.3 Proposed Solution

The proposed solution is an AI-powered Corvit web portal containing an integrated chatbot.

The user enters a question through the chatbot interface. The system processes the query, retrieves relevant information from the Corvit knowledge base, and provides the retrieved context to the AI model. The model then generates a natural-language response.

For questions requiring current external information, online search capability may be used.

Thus, the system combines:

1. Professional web interface.
2. AI conversational assistant.
3. Corvit knowledge base.
4. Retrieval mechanism.
5. GPT-OSS-120B language model.
6. Online search capability.

1.4 Project Objectives

Objective 1 – Intelligent Student Assistance

Develop an AI chatbot capable of understanding natural-language queries and providing relevant Corvit-related information.

Objective 2 – Intelligent Information Retrieval

Integrate a structured knowledge base and retrieval mechanism, with optional online search, to provide relevant and contextual information.

Objective 3 – Professional Web Experience

Develop a responsive and professional web interface that provides simple access to institutional information and chatbot assistance.

1.5 Scope

The system covers:

- Corvit institutional information.
- Course and certification information.
- Course recommendations.
- Timetable and schedule guidance.
- Instructor information.
- Infrastructure information.
- NAVTTC guidance.
- Career guidance.
- AI chatbot interaction.
- Knowledge retrieval.
- Online search when required.
- Responsive website interface.

The system is an assistance platform and does not replace official admission, administrative, financial, or academic decision-making.

2 Overall Description

2.1 Product Perspective

The Corvit AI Chatbot is a web-based system combining a traditional institutional website with an intelligent conversational assistant.

The major layers are:

1. Presentation Layer.
2. Chatbot Interaction Layer.
3. Knowledge Retrieval Layer.
4. AI Processing Layer.
5. Online Search Layer.

2.2 Product Functions

The system provides the following major functions:

- Display Corvit information.
- Accept natural-language questions.
- Retrieve relevant knowledge.
- Generate AI responses.
- Recommend courses.
- Provide timetable information.
- Provide instructor information.
- Provide infrastructure information.
- Provide NAVTTC guidance.
- Provide career guidance.
- Perform online search when necessary.

2.3 User Classes

User	Description
Prospective Student	Explores courses, certifications, and career options.
Current Student	Requires schedules, instructors, facilities, and guidance.
Website Visitor	Seeks general Corvit information.
Administrator	Maintains information and knowledge-base content.

2.4 Operating Environment

The system is intended to operate on:

- Modern desktop and laptop browsers.
- Android and iOS mobile browsers.
- Tablets.
- Internet-connected environments.
- Compatible AI service infrastructure.

2.5 Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript
UI Framework	Tailwind CSS
AI Model	GPT-OSS-120B
Knowledge Base	Structured Corvit data / JSON
Retrieval	RAG-based / semantic retrieval mechanism
Search	Online Search capability
Version Control	Git / GitHub
Deployment	Netlify or compatible hosting

3 Functional Requirements

Functional requirements describe what the system shall do.

Table 1: Functional Requirements

ID	Requirement	Description
FR-01	Chatbot Access	The system shall provide a chatbot accessible from the website.
FR-02	Natural Language Input	Users shall be able to submit questions using natural language.
FR-03	Query Processing	The system shall process user queries before generating responses.
FR-04	AI Response	The system shall generate responses using the configured AI model.
FR-05	Knowledge Retrieval	The system shall retrieve relevant information from the Corvit knowledge base.

FR-06	Context Generation	Retrieved information shall be used as context for response generation.
FR-07	Course Information	The system shall provide available course and training information.
FR-08	Course Recommendation	The system shall recommend potentially suitable courses based on user requirements.
FR-09	Certification	The system shall provide certification-related information when available.
FR-10	Timetable	The system shall provide timetable and schedule information when available.
FR-11	Instructor	The system shall provide instructor information when available.
FR-12	Infrastructure	The system shall provide information about training facilities.
FR-13	NAVTTC	The system shall provide relevant NAVTTC guidance.
FR-14	Career Guidance	The system shall provide general career guidance related to training and certifications.
FR-15	Online Search	The system may use online search for current or external information.
FR-16	Search Processing	Retrieved online information shall be processed before presentation.
FR-17	Conversation Context	The system should preserve active conversation context.
FR-18	Loading Feedback	The interface shall provide feedback while processing a query.
FR-19	Responsive UI	The website shall support desktop, tablet, and mobile layouts.
FR-20	Error Handling	The system shall handle unsupported or invalid queries gracefully.

4 Non-Functional Requirements

4.1 Performance

- The interface should respond without unnecessary delay.
- Chat requests should be processed efficiently.
- Knowledge retrieval should minimize irrelevant information.

4.2 Usability

- The interface shall be simple for non-technical users.
- The chatbot shall be easy to locate and use.
- Controls and text shall be clearly readable.

4.3 Reliability

- The system shall handle invalid queries without crashing.

- The system should provide meaningful fallback responses.
- Temporary service failures should be handled gracefully.

4.4 Scalability

The system architecture should allow additional courses, instructors, information sources, AI models, and future services to be integrated.

4.5 Maintainability

The source code shall be organized so that developers can update the frontend, knowledge base, and AI integration independently where practical.

4.6 Compatibility

The website shall support modern browsers including Chrome, Edge, Firefox, and Safari.

4.7 Security

- API keys shall not be exposed in publicly accessible code.
- User input shall be validated.
- External data shall be processed carefully.
- Sensitive information should not be unnecessarily collected.

5 System Architecture

5.1 High-Level Architecture

The following diagram represents the conceptual architecture.

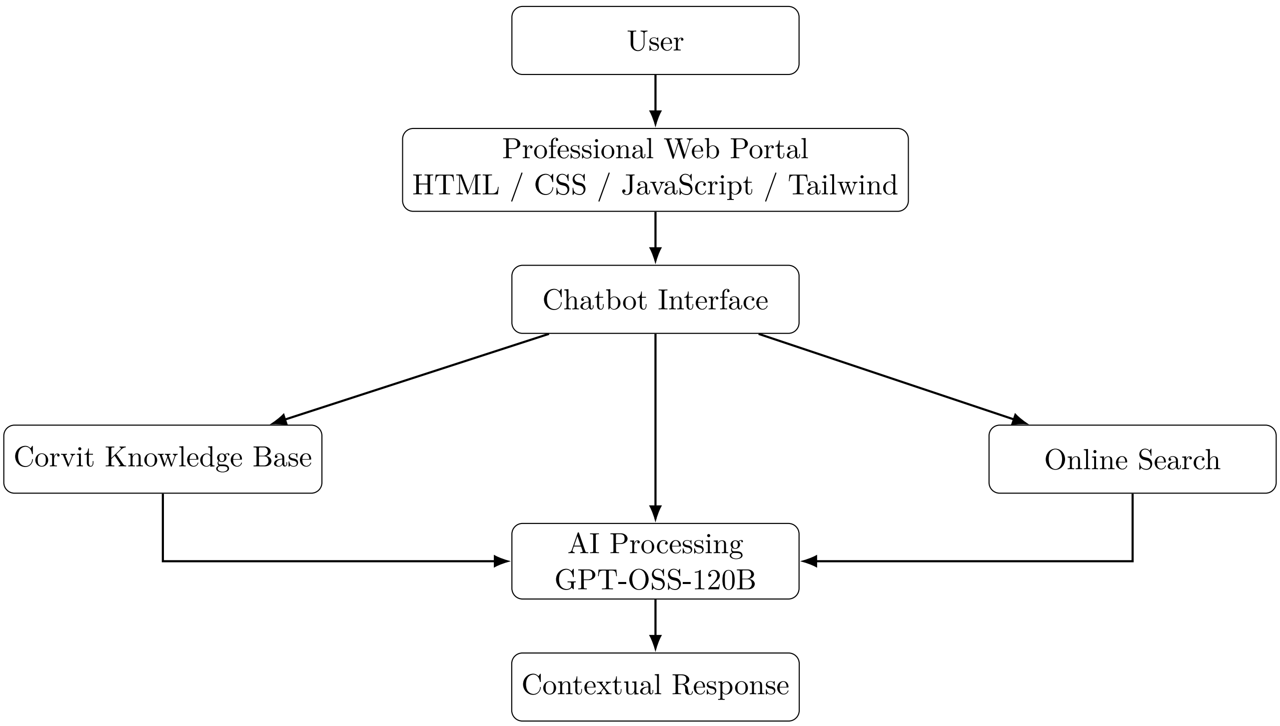

Figure 1: Corvit AI Chatbot High-Level Architecture

5.2 AI Query Workflow

The general processing workflow is:

1. User opens the chatbot.
2. User submits a natural-language query.
3. System analyzes the query.
4. Relevant institutional information is retrieved.
5. Online search may be performed if required.
6. Retrieved information is combined into contextual input.
7. GPT-OSS-120B generates the response.
8. The response is displayed to the user.

5.3 Retrieval-Augmented Generation

The system follows a RAG-oriented architecture in which external knowledge is retrieved before AI response generation.

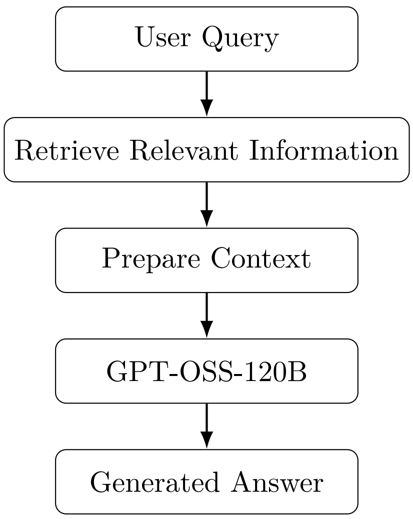

Figure 2: RAG-Oriented AI Processing Workflow

6 Use Case Model

6.1 Major Use Cases

ID	Use Case
UC-01	Ask the AI chatbot a general question.
UC-02	Search or explore available courses.
UC-03	Request a course recommendation.
UC-04	Ask about timetable or schedule.
UC-05	Ask about an instructor.
UC-06	Ask about infrastructure and facilities.
UC-07	Request NAVTTC guidance.
UC-08	Request career guidance.
UC-09	Request information requiring online search.

6.2 UC-01: Ask AI Chatbot

Actor	User
Precondition	User has opened the website.
Main Flow	User opens chatbot → enters query → system processes query → relevant information is retrieved → AI generates response → response is displayed.
Postcondition	User receives a response.

6.3 UC-02: Course Recommendation

Actor	Prospective Student
Precondition	Chatbot is available.
Main Flow	User describes interests or career goal → system retrieves relevant courses → AI analyzes context → recommendations are presented.
Postcondition	User receives course guidance.

6.4 UC-03: Current Information Search

Actor	User
Precondition	Internet connectivity is available.
Main Flow	User asks for current information → system performs online search when applicable → relevant information is processed → AI generates response.
Postcondition	User receives relevant information.

7 Knowledge Base Requirements

7.1 Knowledge Categories

The knowledge base should contain structured information such as:

- Institute profile.
- Courses.
- Certifications.
- Instructors.
- Timetables.
- Infrastructure.
- NAVTTC programs.
- Career guidance.
- Frequently asked questions.

7.2 Conceptual Data Structure

A JSON-based structure may be organized as:

```
{
  "institute": {},
  "courses": [],
  "instructors": [],
  "timetables": [],
```

```
  "infrastructure": [],
  "navttc": {},
  "career_guidance": [],
  "faqs": []
}
```


7.3 Data Quality

Knowledge-base information should be accurate, relevant, structured, consistent, and regularly updated.

8 User Interface and UX Requirements

8.1 Website

The website shall provide a professional and responsive interface containing institutional information and navigation.

8.2 Chatbot

The chatbot should be accessible through a clearly visible button, preferably located in the bottom-right corner of the website.

The chatbot interface should contain:

- Conversation area.
- User input field.
- Send button.
- Loading indicator.
- Message history.
- Error/fallback messages.

8.3 Responsive Design

The interface shall adapt to:

- Desktop screens.
- Laptop screens.
- Tablets.
- Mobile phones.

9 Security and Privacy

9.1 API Protection

AI API credentials shall not be hard-coded into publicly accessible frontend files.

9.2 Input Validation

The system shall validate and sanitize user input before processing.

9.3 Privacy

The system should avoid collecting unnecessary personal information. If user information is stored in future versions, appropriate privacy and access controls should be implemented.

10 Testing Requirements

10.1 Functional Testing

Testing shall verify:

- Chatbot opens correctly.
- Queries are submitted successfully.
- AI responses are generated.
- Knowledge retrieval works.
- Course information is returned.
- Timetable queries work.
- Instructor queries work.
- NAVTTC queries work.
- Career guidance works.
- Online search works when enabled.

10.2 Usability Testing

The system shall be tested for:

- Navigation simplicity.
- Chatbot discoverability.
- Readability.
- Mobile responsiveness.
- User interaction clarity.

10.3 Error Testing

The following cases should be tested:

- Empty query.
- Invalid query.
- Unsupported question.
- Missing knowledge-base information.
- AI service failure.
- Internet failure.

11 Deployment Requirements

11.1 Version Control

The source code shall be maintained using Git and GitHub.

Proposed repository:

`corvit-ai-chatbot`

11.2 Deployment

The professional frontend may be deployed using Netlify or another compatible hosting platform.

The general workflow is:

1. Develop locally.
2. Test the application.
3. Commit source code using Git.
4. Push the repository to GitHub.
5. Connect the repository to hosting.
6. Configure required environment variables.
7. Deploy the website.

12 Constraints and Limitations

The system has several practical limitations:

- AI response quality depends on the selected model.
- Knowledge accuracy depends on the source data.
- Online search requires internet connectivity.
- External AI services may have usage limitations.
- API availability may affect chatbot operation.
- The chatbot should not replace official institutional decisions.

13 Future Enhancements

Future versions may include:

1. Voice-based interaction.
2. Urdu and multilingual support.
3. Student login and personalized assistance.
4. Administrative dashboard.
5. Knowledge-base management interface.
6. Chatbot analytics.
7. WhatsApp integration.
8. Admission assistance.
9. Course registration integration.
10. Document upload and analysis.
11. Advanced semantic vector search.
12. Personalized course recommendation.

14 Conclusion

The Corvit AI Chatbot provides an intelligent solution for improving access to institutional and educational information.

Instead of requiring users to manually search through multiple web pages or repeatedly ask staff common questions, the proposed system provides a conversational interface through which users can request information naturally.

The integration of a structured Corvit knowledge base, retrieval mechanism, GPT-OSS-120B, and optional online search enables the system to provide relevant and contextual responses.

The professional responsive website further improves accessibility and usability. The modular design also allows future integration of voice interaction, multilingual capabilities, personalized recommendations, analytics, admission services, and additional information sources.

15 References

1. IEEE, "IEEE Recommended Practice for Software Requirements Specifications," IEEE Standards Association.
2. World Wide Web Consortium (W3C), "Web Standards and HTML Documentation."
3. Tailwind CSS, "Utility-First CSS Framework Documentation."
4. GitHub, "GitHub Documentation and Git Version Control."
5. Netlify, "Web Deployment and Hosting Documentation."
6. Lewis, P. et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks."

16 Appendix A: Requirement Summary

Category	**Summary**
AI	GPT-OSS-120B based conversational response generation
RAG	Retrieval of relevant Corvit information
Search	Optional online information retrieval
Courses	Course information and recommendations
Students	Timetable, instructor, NAVTTC, and career guidance
Frontend	HTML, CSS, JavaScript, Tailwind CSS
Deployment	Netlify-compatible web hosting
Version Control	GitHub repository